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Def^n: A linear map
$T: X \to Y$ is said to be "bounded below" if $\exists\, c > 0$
such that
$\|T(x)\| \geq c\|x\|$, $\forall x \in X$
↳ obviously if T is bounded below, it wd be injective.

Theorem: Let X, Y be NLS and $T: X \to Y$ be a linear operator.
T is bounded below $\iff$ T is injective
& $T^{-1}: R(T) \to X$ is bounded (continous).
Further, if $\|T(x)\| \geq c\|x\|$
then $\|T^{-1}(y)\| \leq \frac{1}{c}\|y\|$
$\forall y \in R(T)$.

Proof: Suppose T is bounded below,
$\exists\, c > 0$
s.t $\|T(x)\| \geq c\|x\|$ — ①.
Let $x_1, x_2 \in X$ s.t
$Tx_1 = Tx_2$
$\Rightarrow T(x_1 - x_2) = 0$
$\Rightarrow c\|x_1 - x_2\| = 0$ (from ①)
$\Rightarrow x_1 = x_2 \Rightarrow$ T is injective,

Let $y \in R(T)$
$\Rightarrow \exists\, x \in X$ s.t
$Tx = y$
or $T^{-1}y = x$
consider
$\|y\| = \|Tx\| \geq c\|x\|$
$\Rightarrow \|T^{-1}y\| \leq \frac{1}{c}\|y\|$

So T^{-1} is bounded and continous.

converse

Suppose T^{-1} is bounded and injective. (T is)

$\therefore T^{-1}$ is bounded $\rightarrow$ for existence of inverse, we need injectivity of T.

$\Rightarrow \exists\, C>0$ s.t

$\|T^{-1}y\| \leq C\|y\|$

let $x \in X$, $Tx = y$

$\therefore \|x\| \leq C\|Tx\|$

HOMEOMORPHISM

Defn: Let X, Y be a NLS.

A linear map $T: X \to Y$ which is a homeomorphism (i.e bijective and bicontinous) $\rightarrow$ both T and T^{-1} are continous.

is called a linear homeomorphism.

Exercise: Let X, Y be NLS. A bijective linear map from $X \to Y$

i.e $T: X \to Y$ is a homeomorphism.

$\Longleftrightarrow \exists$ positive constants c_1 & c_2 s.T

$$c_1\|x\| \leq \|T(x)\| \leq c_2\|x\| \quad \forall x \in X$$

THEOREM: Let X, Y be NLS and X is finite dimensional. Then every bijective linear map from $X \to Y$ is a homeomorphism.

Further if $\dim X = n$, then X is linearly homeomorphic to $\mathbb{K}^n$

Proof: $T: X \to Y$, bijective linear map

CLAIM: T & T^{-1} are continous

$X \rightarrow$ Finite dimensional.

THEOREM: Let X be a NLS and $f : X \to K$ be a non-zero linear functional with $N(f)$ closed. Then f is continuous, and for any $x_0 \in X \setminus N$

$$\|f\| = \frac{|f(x_0)|}{\operatorname{dist}(x_0, N(f))}$$

Proof: Let $x_0 \in X \setminus N$

$x_0 \notin N(f)$, $f(x_0) \neq 0$.

Let $x \in X$ arbitrary,

$$x = \underbrace{x - \frac{f(x)}{f(x_0)} x_0} + \frac{f(x)}{f(x_0)} x_0.$$

Let $y = x - \frac{f(x)}{f(x_0)} x_0.$

Take $\alpha = \frac{f(x)}{f(x_0)} x_0.$

$\Rightarrow x = y + \alpha x_0.$

consider $f(y) = f(x) - \frac{f(x)}{f(x_0)} f(x_0) = 0.$

$\therefore y \in N(f)$ —— (2)

$$\operatorname{dist}(x, N(f)) = \operatorname{dist}(y + \alpha x_0, N(f))$$
$$= \operatorname{dist}(\alpha x_0, N(f_0))$$
$$= |\alpha| \operatorname{dist}(x_0, N(f))$$
$$= \left|\frac{f(x)}{f(x_0)}\right| \operatorname{dist}(x_0, N(f)).$$

$$\therefore |f(x)| = \frac{|f(x_0)|}{\operatorname{dist}(x_0, N(f))} \operatorname{dist}(x, N(f)).$$

$$|f(x)| \leq \frac{|f(x_0)|}{\operatorname{dist}(x_0, N(f))} \|x\|$$

$\therefore f$ is bounded — (3)

$\& \|f\| \leq \dfrac{|f(x_0)|}{dist(x_0, N(f))}$ — (4)

we need to prove

$\|f\| \geq \dfrac{|f(x_0)|}{dist(x_0, N(f))}$

i.e $|f(x_0)| \leq \|f\| \, dist(x_0, N(f))$.

let $u \in N(f)$.

$|f(x_0)| = |f(x_0) - f(u)|$

$= |f(x_0 - u)|$

$\leq \|f\| \, \|x_0 - u\|$

$\forall u \in N(f)$

Run over all $u \in N(f)$.

$|f(x_0)| \leq \|f\| \, dist(x_0, N(f))$

$\therefore \|f\| \geq \dfrac{|f(x_0)|}{dist(x_0, N(f))}$ — (5).

$\therefore \|f\| = \dfrac{|f(x_0)|}{dist(x_0, N(f))}$.

Note: If (x_n) is a sequence of differentiable functions defined on $[a, b]$ s.t

$x_n(t_0)$ converges to $x(t_0)$, for some $t_0 \in [a, b]$

and $x_n' \longrightarrow y$ as $n \to \infty$ uniformly.

Then $\boxed{x' = y}$

X

$\|$

$\rightarrow$ let $X_0 = C'[0,1] \subseteq C[0,1]$

$Y = C[0,1] = X$

$T : X_0 \longrightarrow Y$

$Tx = x'$

$\rightarrow$ we know T is unbounded.

since X is finite dimensional, so T is bounded.
Note that Y is finite dimensional (basis goes to basis)
$\therefore T^{-1}: Y \to X$ is bounded.

$\Rightarrow$ Also $X \cong \mathbb{K}^n$.

RECALL:

$T \in B(X, Y)$

$N(T)$ or $Ker(T)$ = Null space.

$= \{x \in X \mid Tx = 0\}$.

T is bounded $\Rightarrow N(T)$ is closed.

(Being the preimage of $\{0\}$ under T).

$T^{-1}\{0\} = N(T)$.

Qn: Is the converse true?

$N(T)$ is closed $\overset{?}{\Rightarrow}$ T is bounded

Ans: NO.

EX $X = C^1[0,1]$

$Y = C[0,1]$

$T: X \to Y$ $\qquad x(t)$.

$Tx = x'$

$\Rightarrow T$ is unbounded

$N(T) = \{x \in X \mid x' = 0\}$

= set of constant fn.

→ since $\dim N(T) = 1$,

$N(T)$ is finite dimensional & hence closed.

here $x_n \in X_0$ but $x \in X$

Suppose $x_n \to x$ in X_0 (uniform hence pointwise)

&

$Tx_n \to y$, $y \in Y$ (uniform convergence).

$\overset{\shortparallel}{x_n'}$

$\Rightarrow x' = y$

OR $x \in X_0$ & $Tx = y$.

DEF^n: Let X, Y be NLS and X_0 be a subspace of X. A linear operator $T: X_0 \subseteq X \to Y$ is said to be closed operator if for every sequence (x_n) in X_0, $x_n \to x$ where $x \in X$ & $Tx_n \to y$, $y \in Y$ implies $x \in X_0$ and $Tx = y$.

Example. $T: C^1[0,1] \subseteq C[0,1] \to C[0,1]$

$Tx = x'$ is a closed operator.

Qn $\to$ whether a bounded op. is closed in general?

NO!

$$I: \underset{\uparrow X_0}{C^1[0,1]} \subseteq \underset{\uparrow X}{C[0,1]} \to C[0,1].$$

$Ix = x$

Clearly I is bounded.

Take $x \in C[0,1]$, $x \notin C^1[0,1]$.

$\exists\, x_n \in C^1[0,1]$ s.t (Take polynomials)

$x_n \to x$

$\Rightarrow Ix_n \to x$

But $x \notin C^1[0,1]$.

I is not closed.